

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Classified Practice

Applications of Logarithms

Q001 **Written****Solve** $\log_2 x = 4$.

Q002 Written**Solve** $\log_2(2x - 1) = \log_2 3$.

Classified Practice

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Q003 **Written****Solve all 12 source logarithmic equations, including same-base, product, quotient, and shifted-argument forms.**

Q004 Written**Solve** $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ **completely.**

Classified Practice

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.

Q006 Written**Solve the six source logarithmic inequalities using base monotonicity and argument conditions.**

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Q007 **Written**

Solve all 24 source logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Q008 Written

Solve the source challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

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Q009 Written

Solve $5(\log_2 x)^2 - 16 \log_2 x + 3 < 0$.

Q010 Written

Solve the three source substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.

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Q011 **Written****Solve the source groups: three equations, three inequalities, and three exact logarithmic values.**

Q012 Written**Solve** $(\log_3 x)^2 - \log_3 x - 6 < 0$ **for the positive wave-intensity multiplier** x .

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Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Q014 Written

Find the maximum and minimum of the source logarithmic quadratic on its stated interval, with the corresponding values of x .

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Q015 **Written****Solve the three source extrema problems, each with a different base and interval.**

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

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Q017 **Written**

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Q018 Written

Complete the source digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

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Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Q020 Written**Solve** $\log_x(x + 2) \leq \log_x(4 - x)$ **for all real** x .

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Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Choose the correct option.**A $x < 5$ B $x \geq 5$ C $x > 5$

D all real x

Q022 MCQ**If $\log_2(2x - 1) = \log_2 3$, then $x =$:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

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Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Choose the correct option.**

A 1

B 2

C 4

D 8

Q024 MCQ**For same-base logarithms, equality of logs gives:****Choose the correct option.**

- A arguments equal
- B bases equal
- C arguments add
- D no result

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Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:

Choose the correct option.

A $M > N$

B $M < N$

C $M = N$

D $MN < 0$

Q026 MCQ**For base $0 < a < 1$, the inequality direction is:****Choose the correct option.**

- A preserved
- B reversed
- C removed
- D squared

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Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Choose the correct option.**A $x < 5$ B $x \leq 5$ C $x > 5$ D $x \neq 5$

Q028 MCQ

The warm-up intersection is:**Choose the correct option.**A $0 < x < 5$ B $5 < x < 6$ C $x > 6$ D $-3 < x < 6$

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Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Choose the correct option.**

A $t = x^2$

B $t = \log_a x$

C $t = a^x$

D $t = 1/x$

Q030 MCQ**The inequality $5t^2 - 16t + 3 < 0$ gives:****Choose the correct option.****A** $t < 1/5$ **B** $1/5 < t < 3$ **C** $t > 3$ **D** $0 < t < 1$

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Q031 MCQ

The substitution variable must satisfy:**Choose the correct option.**A $t > 0$ B $t \in \mathbb{R}$ C $t < 0$ D $t = 1$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:

Choose the correct option.

A $x < 1/9$

B $1/9 < x < 27$

C $x > 27$

D $0 < x < 3$

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Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Choose the correct option.

A $t = 0$

B $t = 1$

C $t = 2$

D $t = 3$

Q034 MCQ

The correct first step for extrema is:**Choose the correct option.**

- A differentiate in x
- B set $t = \log_a x$
- C ignore the interval
- D take $x = 0$

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Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Choose the correct option.**A $[-1,3]$ B $[0,3]$ C $[1,27]$

D all real

Q036 MCQ**The water-tank challenge maximum health rating is:****Choose the correct option.****A** 3**B** 4**C** 5**D** 8

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Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Choose the correct option.**

A $\log_{10} N$

B $\lfloor \log_{10} N \rfloor + 1$

C $N+1$

D 10^N

Q038 MCQ**If $-9 < \log_{10} N < -8$, the first non-zero digit is at:****Choose the correct option.**

- A 8th
- B 9th
- C 10th
- D 11th

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Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Choose the correct option.**

A 2

B e

C 10

D any base

Q040 MCQ**The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:****Choose the correct option.****A** (0,1)**B** (1,4)**C** (0,4)**D** empty

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Applications of Logarithms

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $\log_5(x - 1) = 2$.

Choose the correct option.

A $x = 26$

B 0

C 1

D cannot be determined

Quiz MCQ 2 Concept Quiz · MCQ

For $0 < a < 1$ and $M < N$, compare $\log_a M$ and $\log_a N$.

Choose the correct option.

A $\log_a M > \log_a N$

B 0

C 1

D cannot be determined

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Applications of Logarithms

Quiz MCQ 3 Concept Quiz · MCQ

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.**Choose the correct option.**A $x = 2, \sqrt{2}$

B 0

C 1

D cannot be determined

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2 \log_3 x + 4$ on $1 \leq x \leq 27$.

Choose the correct option.

A 5 at $x = 3$

B 0

C 1

D cannot be determined

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Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\log_3(x + 1) = 2$.

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.

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Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written

Solve $(\log_5 x)^2 - 4 < 0$.

Quiz WRITTEN 8 Concept Quiz · Written

How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?

Answer key

Use the question number shown on each page.

- Q001 16
- Q002 $x = 2$
- Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
- Q004 $x = 4$
- Q005 $0 < x < 81$; $5 < x < 6$
- Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
- Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
- Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
- Q009 $2^{1/5} < x < 8$
- Q010 Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
- Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
- Q012 $1/9 < x < 27$
- Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
- Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
- Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
- Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
- Q017 24 digits; 9th decimal place.
- Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
- Q019 55 digits; leading digit 1.
- Q020 No real solution.
- Q021 C
- Q022 B
- Q023 C
- Q024 A
- Q025 B
- Q026 B
- Q027 C
- Q028 B
- Q029 B
- Q030 B
- Q031 B
- Q032 B
- Q033 C
- Q034 B
- Q035 B
- Q036 C
- Q037 B
- Q038 B
- Q039 C
- Q040 D
- Quiz MCQ 1 A
- Quiz MCQ 2 B
- Quiz MCQ 3 C
- Quiz MCQ 4 D
- Quiz WRITTEN 5 $x = 8$
- Quiz WRITTEN 6 $x \geq 8$
- Quiz WRITTEN 7 $1/25 < x < 25$
- Quiz WRITTEN 8 16 digits